

Project Title: Uncovering Trust and Transparency Issues in Auto Repair

Group ID: 6

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Project Summary:

This project applies data science techniques to analyze customer reviews of auto repair shops in the Bay Area in order to identify common complaints and consumer pain points within the automotive services industry. Online platforms such as Yelp and Google Reviews contain large volumes of unstructured text that capture real customer experiences, yet these data are rarely analyzed in a systematic and data-driven manner. Auto repair services are particularly challenging for consumers due to pricing uncertainty, communication gaps, and variability in service quality, making this domain well suited for analytical investigation. Using the Yelp Open Dataset and Google Reviews, we will collect review text, star ratings, and business metadata for auto repair shops across the Bay Area. After data cleaning and preprocessing, we will conduct exploratory data analysis to examine rating distributions, temporal trends, and geographic patterns. Natural language processing techniques, including sentiment analysis and topic modeling, will then be applied to identify recurring themes in customer feedback and to assess how specific issues influence overall satisfaction. We expect the analysis to reveal a small number of dominant complaint categories, such as pricing transparency, repair quality, and customer communication, that are strongly associated with negative sentiment and lower ratings. The results will demonstrate how unstructured review data can be transformed into actionable insights, benefiting consumers, informing service providers, and highlighting the real-world applicability of data science methods in understanding customer behavior and improving service-oriented industries.

- **Broader Impacts**

This project has broader significance for consumers, service providers, and society by addressing long-standing transparency and trust issues within the automotive repair industry. For consumers, particularly everyday drivers who rely on repair services but often lack technical knowledge, the findings can provide clearer insight into common sources of dissatisfaction, such as unclear pricing, poor communication, and inconsistent service quality. Greater awareness of these patterns may empower customers to make more informed decisions and advocate for clearer explanations and fairer pricing. Service providers, including independent repair shops and dealerships, may also benefit from the analysis by gaining a data-driven understanding of customer expectations and recurring pain points. Identifying which issues most strongly influence negative sentiment can help businesses improve communication practices, pricing transparency, and overall service quality, ultimately strengthening customer trust and long-term relationships. From a broader societal perspective, this project demonstrates how large-scale, unstructured review data can be leveraged to uncover systemic issues in essential service industries. The results may be of interest to consumer advocacy organizations, policymakers, and technology developers seeking to improve fairness and accountability in automotive services. More broadly, the project highlights the role of data science in reducing information asymmetry, improving consumer protection, and supporting evidence-based improvements in service-oriented markets that affect millions of people on a daily basis.

- **Data Sources**

1. **Yelp Open Dataset:** <https://business.yelp.com/data/resources/open-dataset/>

Provides customer review text, star ratings, business categories, timestamps, and location data for automotive repair shops in the Bay Area. This dataset will be used to analyze post-service customer satisfaction and recurring complaints.

2. Reddit Automotive Repair Communities

These data include user posts and comments discussing repair experiences, pricing concerns, complaints, and advice-seeking behavior, capturing consumer uncertainty prior to repairs. Collected using the Reddit API.

- Expected Major Findings / Project Questions

1. Pricing transparency is a primary driver of negative sentiment. We hypothesize that complaints related to unclear or unexpected pricing will be the most frequent themes in negative Yelp reviews and Reddit discussions.
2. Communication quality strongly influences customer satisfaction. We expect reviews that mention poor explanations, lack of updates or unclear diagnoses to correlate with lower star ratings and more negative sentiment.
3. A small number of complaint categories dominate negative feedback Topic modeling is expected to reveal a limited set of recurring issues such as pricing repair quality and trust that account for the majority of negative sentiment.